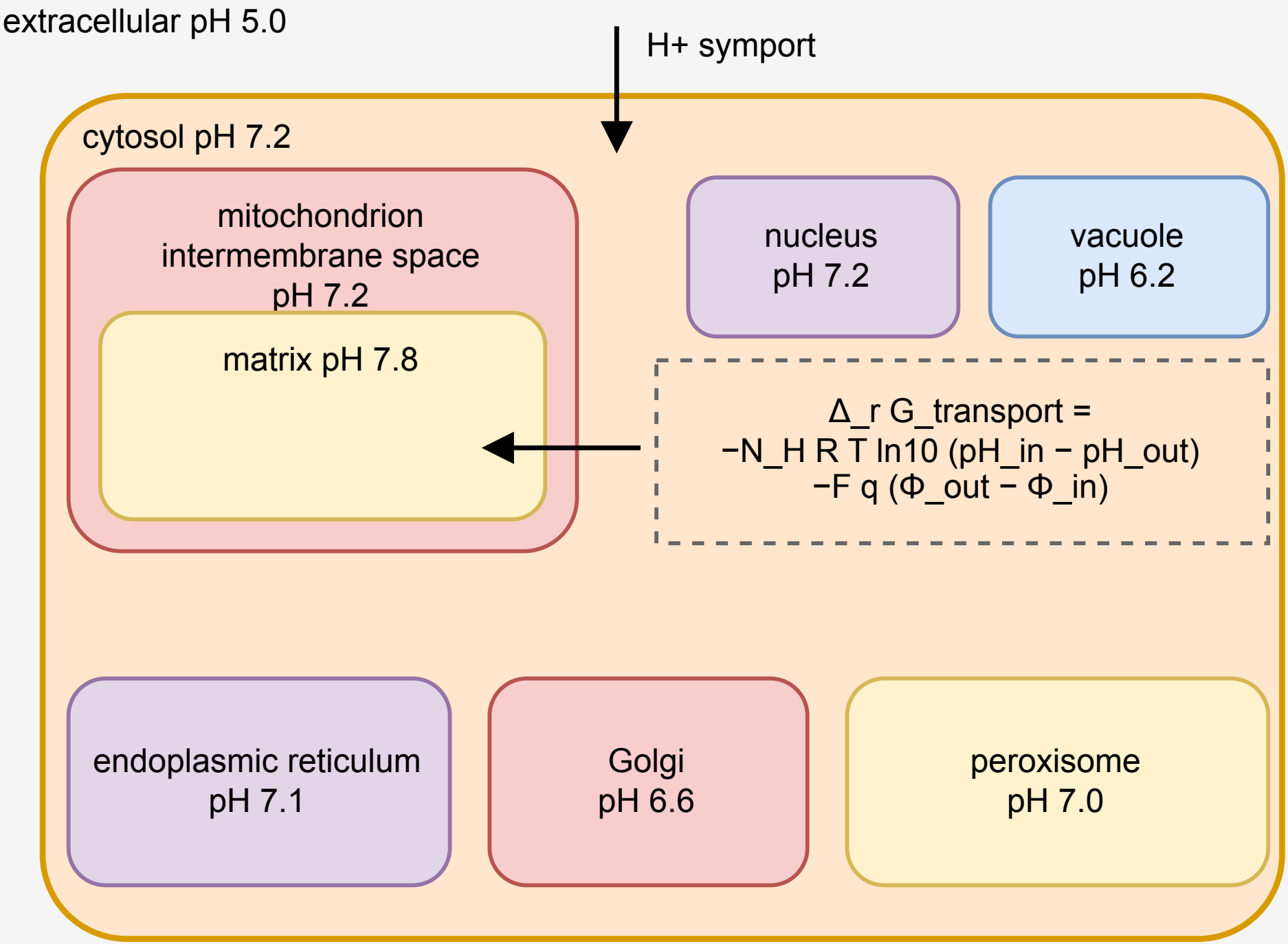
extracellular pH 5.0
H+ symport
cytosol pH 7.2
mitochondrion
intermembrane space
pH 7.2
matrix pH 7.8
nucleus
pH 7.2
vacuole
pH 6.2
Δ_r G_transport =
−N_H R T ln10 (pH_in − pH_out)
−F q (Φ_out − Φ_in)
endoplasmic reticulum
pH 7.1
Golgi
pH 6.6
peroxisome
pH 7.0